



FACULTY OF TECHNOLOGY
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UK Visas & Immigration
Global Talent Visa – Tech Nation (Digital Technology)
United Kingdom

To Whom It May Concern

I write to endorse Odeyemi Olusegun Israel for the UK Global Talent Visa under the Digital Technology route. I am a Professor of Electrical and Electronic Engineering at the University of Ibadan, where I currently serve as Provost of the Postgraduate School. My research spans power systems, computational intelligence, and applied machine learning. I have mentored Olusegun for over 10 years and have reviewed his technical work in detail.

AMTTP: A Production-Grade Compliance Platform Built by One Individual

Olusegun has designed and built the Anti-Money Laundering Transaction Trust Protocol (AMTTP), a highly advanced, production-grade compliance platform at proof-of-concept stage, developed entirely by one individual.

The system comprises a four-layer architecture including client SDKs (TypeScript and Python, both open-sourced), a compliance orchestration engine of seven parallel microservices, an ML training pipeline processing 2.64 million Ethereum transactions (achieving a production-level ROC-AUC of 0.9879 in testing environments), and 18 Solidity smart contracts deployed on Ethereum Sepolia. The entire stack operates across 17 containerised microservices.

AMTTP is designed to address a key limitation in the blockchain compliance industry: most existing tools, including Chainalysis, Elliptic, and TRM Labs, primarily operate after a transaction has settled. In contrast, AMTTP introduces a pre-settlement compliance architecture, where machine learning inference, graph analysis, screening sanctions, and policy evaluation are performed within Ethereum's 12-second block window. This represents a meaningful advancement toward real-time, execution-aware compliance systems.

He also developed zkNAF, a zero-knowledge proof framework (Groth16 ZK-SNARKs on BN254) that allows institutions to prove compliance facts—such as KYC validity, risk-score range, and sanctions non-membership—without revealing sensitive personal data. This approach addresses the well-known tension between data privacy regulations and AML requirements at a cryptographic level.

In a university environment, a project of this scope would typically require a principal investigator, multiple postdoctoral researchers, and a team of PhD students. Olusegun has designed and implemented this system independently. I have reviewed the GitHub repository, and there is no evidence of external collaboration or outsourced development.

Blind-Spot Decomposition and System Analysis

Olusegun's research papers on Blind Spot Decomposition Theory (BSDT) and the Universal Detection Principle (UDP) demonstrate strong capability in anomaly detection. However, I particularly highlight his work on Blind-Spot Decomposition and the Geometry of System Collapse, which aligns closely with my expertise in power systems.

In the case of the February 2021 Winter Storm Uri event affecting the Texas ERCOT grid, his adaptive friction framework, applied to dispatch data, achieves:

- Detection approximately 72 hours before rolling blackouts began (AUC = 0.9997)
- A reduction in peak capacity loss of up to 71% under adaptive control
- Identification of gas supply chain disruption as the primary systemic vulnerability

This aligns with findings from the subsequent FERC/NERC investigation, but notably, his framework derives this insight mathematically in advance using system data alone. From a power systems perspective, achieving this level of predictive accuracy with meaningful lead time is highly significant.

More broadly, this work demonstrates his ability to translate theoretical models into practical, data-driven systems—an ability that is clearly reflected in the design and implementation of AMTTP.

Relevance to UK and Future Potential

The UK's evolving regulatory landscape for digital assets, particularly under the FCA's forthcoming crypto-asset framework, is expected to increase demand for infrastructure that supports real-time, execution-aware compliance. AMTTP's architecture is well aligned with these emerging needs.

While the platform is currently at a proof-of-concept stage, its technical depth, system completeness, and performance characteristics indicate strong potential for real-world deployment within institutional blockchain and compliance environments in the UK.

In addition, Olusegun's interdisciplinary work—spanning blockchain systems, machine learning, and applied mathematics—aligns with UK priorities in financial technology, infrastructure resilience, and advanced computational research.

Final Recommendation

In my years at the University of Ibadan, I have mentored hundreds of engineers. Olusegun's engineering capability and systems-level thinking place him among the most capable individuals I have encountered. He demonstrates the rare ability to design and implement complex, interdisciplinary systems from first principles across machine learning, blockchain, cryptography, and applied mathematics, entirely without institutional support.

I believe he represents a strong example of emerging technical leadership, with clear potential to contribute meaningfully to the UK's digital technology ecosystem.

I recommend him for the Global Talent Visa under the Exceptional Promise category without reservation.

Yours sincerely



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